A proof of Maier's Bessel-number formula

Principia Math in Progress

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Task 1. Prove the generalized Eulerian identity. Write $(a)_n = a(a+1)\cdots(a+n-1)$ and $(a)_{n,2} = a(a+2)\cdots(a+2n-2)$, with both empty products equal to one. The generalized Stirling numbers satisfy

$$S_{n+1,j}(a,b;r) = (-an+bj+r)S_{n,j}(a,b;r) + S_{n,j-1}(a,b;r), \qquad S_{0,0}=1,$$

and vanish outside $0 \le j \le n$. Put $B_{n,j} = S_{n,j}(1,2;0)$ and $B^{(r)}_{n,j} = S_{n,j}(1,2;r)$. The generalized Eulerian numbers $E_{n,k}(a,b;c_0,c_\infty)$ are similarly normalized and satisfy

$$E_{n+1,k} = (-an+bk+c_0)E_{n,k} + ((a+b)n - b(k-1) + c_\infty)E_{n,k-1}. \tag{1}$$

We prove Conjecture 4.20 of [1], including all its parameters.

Theorem 1. *Let $p \ge 0$ be an integer, $\zeta \in \{0,1\}$, and $m = 2p+\zeta$. For $0 \le k \le n$,*

$$E_{n,k}(-1,2;c+m,m) = \frac{(c+m)_n}{(c+2p+2\zeta)_{p,2}} \sum_{\ell=0}^{p} \frac{(c+1)_{p,2}}{(c+1)_{k+\ell,2}} B_{m,m-\ell} B^{(m-2\ell)}_{n,n-k}. \tag{2}$$

The equality holds as a rational-function identity in c. Its right side has removable singularities wherever its displayed denominators vanish, with polynomial continuation equal to the left side.

Proof. We first work at generic c, so every denominator used below is nonzero. The Stirling recurrence, or direct coefficient comparison, gives

$$\sum_{n \ge j} B^{(r)}_{n,j} \frac{z^n}{n!} = \frac{(1+z)^r}{j!}(z+z^2/2)^j.$$

Consequently

$$B_{m,m-\ell} = \frac{m!}{2^\ell \ell!(m-2\ell)!}, \qquad B^{(r)}_{n,n-k} = \frac{n!}{(n-k)!}[z^k](1+z)^r(1+z/2)^{n-k}. \tag{3}$$

Let

$$P_k(y) = \sum_{\ell=0}^{p} \frac{B_{m,m-\ell}}{(c+1)_{k+\ell,2}} y^{m-2\ell}, \quad H_{n,k} = \frac{n!}{(n-k)!}[z^k](1+z/2)^{n-k}P_k(1+z), \quad q = \frac{(c+1)_{p,2}}{(c+2p+2\zeta)_{p,2}}. \tag{4}$$

Define $H_{n,k} = 0$ outside $0 \le k \le n$. The proposed right side of (2) is $F_{n,k} = q(c+m)_n H_{n,k}$.

We record two polynomial identities. Set $\alpha = c+2k+m-1$, $\beta = 2m+c+2k-2$, and $\gamma = m\alpha$. Then

$$\mathcal{D}_k P_k := (y^2-1)P_k'' - \beta y P_k' + \gamma P_k = 0, \tag{5}$$

and, when $k \ge 1$,

$$P_{k-1} = \alpha P_k - y P_k'. \tag{6}$$

To verify (5), write the coefficient of $y^{m-2\ell}$ in P_k as a_ℓ. Formula (3) gives

$$2(\ell+1)(c+2k+2\ell+1)a_{\ell+1} = (m-2\ell)(m-2\ell-1)a_\ell.$$

These are exactly the coefficient equations in (5); the top and terminal terms also vanish because $m-2p \in \{0,1\}$. For (6), the coefficient multiplier is $\alpha - (m-2\ell) = c+2k+2\ell-1$, the ratio of the two step-two denominators.

We now show

$$(n+c+m)H_{n+1,k} = (n+2k+c+m)H_{n,k} + (n-2k+2+m)H_{n,k-1}. \tag{7}$$

For $k=0$, this is immediate from $H_{n,0} = P_0(1)$. For $1 \le k \le n+1$, put

$$N = n-k+1 \ge 0, \quad A = n+c+m, \quad d = n-2k+2+m, \quad U = 1+z/2, \quad y = 1+z.$$

A coefficient calculation using (4) and (6) gives

$$\begin{aligned} \frac{N!}{n!}\{AH_{n+1,k} - (A+2k)H_{n,k} - dH_{n,k-1}\} \\ = -[z^{k-1}]U^N(\mathcal{D}_k P_k)(1+z) = 0. \end{aligned} \tag{8}$$

Here is an explicit verification of the calculation, to keep (8) self-contained. For any formal series h, P,

$$\begin{aligned} [z^{k-1}]hP' &= k[z^k]hP - [z^{k-1}]h'P, \\ [z^{k-1}]hP'' &= k(k+1)[z^{k+1}]hP - 2k[z^k]h'P + [z^{k-1}]h''P. \end{aligned}$$

Substituting $P_{k-1} = \alpha P - yP'$, the first expression in (8) becomes $[z^k]U^{N-1}KP$, where

$$K = A(N+k)U - (A+2k)N - d\alpha zU + dkUy - dz(Ny/2+U).$$

Moving derivatives off P in $[z^{k-1}]U^N\mathcal{D}_k P$ gives $[z^k]U^{N-1}LP$, where

$$\begin{aligned} L = {} & 2k(k-1)U^2 - 2k(N+1)zU + 2(N+1)zU + \tfrac{1}{2}N(N+1)z^2 \\ & - \beta kUy + \beta z(Ny/2+U) + \gamma zU. \end{aligned}$$

In these expressions $A = N+k-1+c+m$ and $d = N-k+1+m$. Expansion gives $K+L=0$, proving (8). This verification also covers $N=0$, or $k=n+1$: the $H_{n,k}$ term has multiplier $N=0$, while U^{-1} is a well-defined formal series. Thus no undefined factorial is used at this edge. Equation (5) proves (7).

It remains to check the apex normalization. Using (3),

$$\begin{aligned} P_0(1) &= \sum_{\ell=0}^{p} \frac{(-p)_\ell\,(\frac{1}{2}-p-\zeta)_\ell}{((c+1)/2)_\ell\,\ell!} \\ &= \frac{(c/2+p+\zeta)_p}{((c+1)/2)_p} = q^{-1}. \end{aligned} \tag{9}$$

The middle equality is the finite Chu–Vandermonde identity. It follows, for example, by expanding $(1-t)^p$ in the beta integral and then continuing the resulting rational identity in its parameters. Hence $F_{0,0} = 1$. Multiplying (7) by $q(c+m)_n$ shows that F satisfies (1) with $a=-1, b=2, c_0 = c+m, c_\infty = m$. The triangular recurrence and its apex determine every entry uniquely, so $F = E$ by induction on n. Finally, this equality at generic c is an identity of rational functions. The recurrence makes E a polynomial in c, proving the claimed cancellation and continuation at every exceptional value. □

Task 2. Find a combinatorial interpretation. For parameter values that make all terms nonnegative, explain (2) by a weighted counting model and interpret the summation index ℓ.

Task 3. Study the auxiliary polynomial family. Identify P_k within classical polynomial families, and use its differential equation to obtain useful connection formulas or zero information.

Task 4. Extend the parameter range of finite formulas. Determine which additional pairs (a, b) admit analogous finite Bessel-type expansions for $E_{n,k}(a, b; c_0, c_\infty)$.

Task 5. Remove apparent singularities explicitly. Derive formulas with manifest polynomial dependence on c, including efficient evaluations at negative integer and negative odd integer values.

References

[1] R. S. Maier, *Triangular recurrences, generalized Eulerian numbers, and related number triangles*, *Advances in Applied Mathematics* **146** (2023), 102485. Accepted manuscript, arXiv:2207.10224v2, Definition 4.4, Definition 4.6, and Conjecture 4.20 (page 44).